CellForge: Language-Model Agents as Direct Authors of Neural Networks

Authoring, breeding and evolving small gradient-free cell graphs

CellForge project*

Living draft, revision 4, 8 October 2026

Abstract

We ask whether large language model (LLM) agents can build useful neural computation by *directly writing* weights, biases and wiring, with no gradient-based fitting of the resulting artifact, and whether small authored cells can then be recombined by breeding and evolution. CellForge provides a recursive graph representation (primitive cells composed into nested, shareable bundles with optional conditional dispatch), atomic validated transactions, a typed `import_def` crossover, an append-only hash-chained ledger, and a final evaluation on a separate machine with freshly generated cases. On variable-length digit strings, four independently authored artifacts—including one written with no metric feedback—read in-distribution final cases perfectly (exact match 1.000, $n = 2000$); under distribution shift the best reads 73.8% of strings in an unseen font (72.3% under size shift; the others 38–57%), against 10.0% for a gradient-trained slot-MLP control (83.8% in distribution) and 0% for random direct edits. Authored character language models reach 0.678–0.681 bits per character on a synthetic grammar, but authored and control models score near chance (0.224–0.266) on public multiple-choice probes. On ARC-AGI-2 public-training development tasks, authored graph cells solve 3/200 tasks and 4/200 with one object gene; breeding, mutation and random search mostly tie, and an evolved cellular-automaton control solves 0/40. These are development results on exposed data, from single small runs, with no private-set score. A later four-author literal-tensor batch reaches 1/4 search and 0/4 development tasks, matching a hand-coded baseline. Separately, direct per-task Codex calls solve 4/4 previously exposed search tasks and 7/7 newly selected public-training tasks. Across 12 planned tasks, 11 were scored and one was cancelled before author creation. The 11/11 scored result is a small, nonrepresentative sample with unverified model prior exposure, not a benchmark score or a literal-weight result. On a separate synthetic spatial rule family, two founders reach 24/24 held-out cases but implement one functional behavior. A two-specialist bank then solves 24/24 fresh synthetic tasks versus 12/24 for either specialist; its authored router is exactly fixed identity routing, which also solves 24/24. These results show bounded parameter transfer and composition within host-written algorithms, not learned-router superiority, sustained evolution, or general intelligence. Two fixed-T8 studies yield identity-duplicate founders (0/17 exposed training demonstrations), a mutation timeout and an explicit-objective founder timeout. Missing candidates and uncomputed paired randoms leave both two-attempt numerical comparisons incomplete; the prompt-wording effect is unestimated. Dispatcher checks are operational evidence only. Exact native-author credit use remains unknown, and the full literal-weight qualification gates remain unsuccessful.

*Historical orchestration and maker agents: Claude (Anthropic) in Claude Code. Later native-author provenance is reported per experiment; it includes gpt-6-luna where recorded. Project owner: [name to be supplied].

1 Introduction

Neural networks are almost always *fitted*: an optimizer adjusts parameters to reduce a loss on data. LLMs, however, can read images, reason about mechanisms and write precise numerical specifications. This suggests a different construction process: an LLM *authors* a network the way an engineer writes a program, observes its failures and revises it. CellForge is an experimental system for testing this idea, and for testing whether small authored modules (*cells*), grouped into *bundles* and *circuits*, can be joined by *breeding* (typed crossover of imported definitions) and *evolution* (discrete mutation and selection). Everything is pure in the sense of the project owner's rule: no gradients of any kind, including fitness-weighted gradient estimators, are used in primary arms. The target hardware is low power: two CPU cores and 4 GB of RAM.

Our primary question is whether LLM agents can discover useful neural computations, their recursive organization and better ways of developing them, using evidence-backed memory and direct numerical and structural edits, producing an offline artifact that improves on independent evaluation.

How results are validated. In the early experiments, development data, reference anchors and all diagnostics are author-visible and count as development feedback. Final cases are rendered on a separate server by an `evaluator` account with a fresh secret seed per evaluation, and only aggregates leave the machine. Every run records hashes, ledger entries and a receipt file under `runs/`; each claim below names its receipt where one exists. Many later experiments use exposed public development tasks for which no final split exists; those are labelled development results throughout. Section 5 separately reports later local experiments whose final synthetic panels were procedurally sealed until search selection; these were not separate-server or adversarially audited evaluations. This revision integrates the specified October reports and preserves historical results; it is not a new audit of every server campaign in `LAB.md`.

Scope and non-goals. The artifacts studied are an offline image-to-text recognizer, small authored byte-level language models, and authored grid-transformation graphs. None has language understanding, dialogue, speech capability or general intelligence. The project owner's long-term aim is more ambitious (a conversational, multimodal system whose computation is constructed rather than fitted); we state it only to explain design choices (modality-agnostic decoder, recursive reusable components) and keep it strictly separate from our claims. Section 7 lists what would have to be demonstrated, in order.

Contributions. (1) A recursive cell/bundle IR with lazy conditional dispatch, content hashing, behaviour-preserving refactors and validity-preserving typed crossover; (2) a transaction and governance design that makes each parameter change traceable to an accepted proposal and separates development feedback from independent evaluation; (3) measurements of LLM-authored artifacts against gradient-trained and random-search controls on digit strings, synthetic and real text, MNIST-1D, public multiple-choice probes, and ARC-AGI-2 development tasks, including several negative results; (4) a later literal-tensor study distinguishing an ARC development null, synthetic rule transfer, identity-router composition, a bounded shared-NCA null with a terminal author timeout, and operational dispatch checks. These are separate evidence classes, not successive improvements on one benchmark.

2 Related work

Architecture search composes motifs hierarchically [1] and evolves topologies and components [2]; both fit weights by training. Our evidence memory adapts ideas from visual retrieval-augmented generation [3, 4, 5]. Neural cellular automata have been applied to ARC-AGI with a separate gradient-trained model per task [6]; our evolved variant (Section 4.9) replaces gradients by discrete selection and is a different experiment. Tracr compiles hand-written programs into transformer weights without gradient descent [7], the closest prior example of directly constructed weights that we recorded. A recorded report of LLM-seeded genetic search over fixed-width programs on ARC-AGI-2 [8] motivates our validity-preserving genome; we did not independently verify its numbers and do not compare against them. We are not aware of prior work in which LLMs author all weights of a working recognizer without fitting, but a wider prior-art review is required before any novelty claim.

ARC harness context (primary sources checked 6 October 2026). ARC Prize publishes MIT-licensed benchmarking harnesses for ARC-AGI-1/2 and ARC-AGI-3 [9, 10]. Its Astra results list best observed ARC-AGI-1 accuracy of 98.5% (high or xhigh) and ARC-AGI-2 accuracy of 95.0% (max). For ARC-AGI-3 semi-private, the Provider Adapter reaches 99.95% at high reasoning ($18,817 total), versus 62.71% at max reasoning ($26,098 total) with Standard [11, 12]. These are different reasoning settings and harnesses, not a controlled estimate of the harness effect alone. The Standard interface retains model-selected notes; the Provider Adapter also retains opaque reasoning state and uses compaction [12]. ARC-AGI-3 reports action efficiency, not percent of games solved; its percentages must not be substituted for our level counts or exact-grid task accuracy. Benchmark near-saturation is not proof of AGI [12].

Poetiq's MIT-licensed solver executes proposed programs on demonstrations and builds subsequent prompts from their errors and scores [13]. Its December 2025 report records an ARC Prize-verified 54% on ARC-AGI-2 semi-private at $30.57 per problem [14]; this is a dated result, not a current ranking claim. Tycho's Apache-2.0 repository describes executable world-model construction, checking and planning, and reports 100.00 RHAE on 25 public ARC-AGI-3 games with GPT-5.6 Sol and Claude Opus 5 [15]. The community leaderboard classifies such public scores as self-reported unless otherwise indicated [16]. Public and verified semi-private scores are not interchangeable. These systems motivate explicit memory, feedback and verification ablations; they do not demonstrate CellForge's literal-weight learning or evolutionary mechanism.

Literature work still pending. *[L1: Pin exact harness/solver commits, configurations and model versions before any reproduction.] [L2: Verify the ARC-AGI-2 Astra cost row and its unit from a stable primary receipt; no per-task price is asserted here.] [L3: Complete the primary-source comparison of program refinement, world-model verification and literal-weight mutation, including exposure and compute matching.]* No external solver was installed or run for this literature update.

2.1 Automated research and language-model reward design

The following comparison distinguishes the artifact an LLM writes from the mechanism that changes policy weights. These studies inform experimental design; none establishes a CellForge result or demonstrates its literal-weight population standard.

Autoresearch as training-code search. Karpathy's `autoresearch` permits edits to `train.py`, while `program.md` fixes the data/evaluation code in `prepare.py` [18]. Each trial has five minutes

of training on one GPU, excluding startup/compilation, and the agent retains changes according to validation bits per byte. The inspected training implementation uses backpropagation with Muon/AdamW. The LLM therefore searches training programs and settings; an optimizer fits model weights. Repeatedly inspected validation scores are search feedback, not an untouched final test. A per-trial time limit alone does not equate total search effort, hardware, agent tokens or retries. The cited branch was inspected on 8 October 2026; its commit remains unpinned.

ENPIRE as a physical feedback harness. ENPIRE first constructs safety, reset and verification interfaces with human feedback, then freezes them for policy improvement [19]. The second stage supports heuristics, generated policy code, behavior cloning and reinforcement learning. Its headline 99% success must be read under the paper's eight-retry protocol, which permits recovery informed by earlier failure; it is not pass@1 or independent best-of-eight sampling. Appendix A.2 describes an LLM-written binary verifier from labeled success/failure examples, constrained by accuracy and latency requirements. The agent cannot change metric computation but may inspect failures; that feedback exposure must remain explicit when interpreting the described held-out set. The relevant transfer to CellForge is a fixed interface and independently checked success signal, not a claim that the LLM directly authors all policy tensors. The paper and official code repository are separate cited sources; no robot experiment was reproduced here [20].

Eureka as reward-program search. Eureka generates and revises reward code from environment descriptions and training feedback; PPO supplies the policy-weight updates [21]. It reports outperforming human rewards on 83% of 29 environments. That is a task-wise comparison under its search/selection protocol, not single-proposal reliability. The method uses five independent search restarts, each with five iterations of sixteen proposals. Intermediate rewards receive one PPO run, with acknowledged noise; final rewards receive five runs. Main results select the best search outcomes, and policy evaluation uses maximum-checkpoint metrics. These selection and repetition details matter when comparing an evolutionary author against random edits under a finite budget.

Text2Reward as dense shaping code. Text2Reward maps instructions and a Pythonic environment abstraction to executable dense reward code, then trains policies using PPO or SAC [22]. It reports comparable or better success and convergence than expert rewards on 13 of 17 manipulation tasks. Execution feedback repairs code errors, while human feedback can refine intended behavior. Perception is supplied by other components, and the paper notes that symbolic rewards may omit relevant aspects of the goal. Thus a valid program need not capture user intent, and reward synthesis remains distinct from directly writing policy weights.

Implications for the proposed CellForge study. Our methodological inference is to freeze the task-exact evaluator, exposure partitions and control semantics before search. Any proposed shaping objective should be recorded separately from final success, and must not change protected tests or turn invalid control execution into evidence of improvement. Account for the entire search, including rejected proposals, retries and independent seed repetitions, rather than comparing only winning artifacts. Preserve first-attempt accuracy alongside any retry-assisted result and freeze selection before final access. For the literal-weight claim, keep host source fixed and expose only permitted training failures to the author; separately label any reward-code or algorithm search. These are proposed safeguards, not imported empirical gains, execution authorization or permission to reopen closed campaigns.

3 System

3.1 Representation

The following graph and operator descriptions document the early experimental protocols. The later literal-weight standard requires LLM-authored founders and mutations, joint cell/router tensors, matched charged random edits, crossover, and improvement across generations. Early gene selection, count-fitted seeds and automated numeric search remain baselines under that standard; their presence in this historical paper does not qualify them as the current primary experiment. Later spatial cells use tensors to parameterize host-written algorithms, so "weight" does not by itself imply a learned neural implementation of those algorithms.

A *cell* computes $h = \sigma(W_1 x + b_1)$, $y = W_2 h + b_2$ over the last axis of its input. A *bundle* is a typed DAG whose nodes are parameter-free operations, parameter tensors, cells, other bundles, or a top-1 `dispatch` among bundles; unselected branches do not execute. The same representation is used at every depth; definitions may be shared and parameter sharing is explicit. Recursive template references and exponential expansion are rejected at validation. A model has an `encode` bundle (pixels $\rightarrow$ memory) and a `step` bundle (memory, step index, previous token, state $\rightarrow$ logits, state), driven by a bounded greedy decoder that stops on the model's own EOS. Structural profiles (ladder, ladder-micro, open) constrain organization and are checked by a validator.

3.2 Construction protocol and maker operator

Makers submit JSON proposals of atomic operations (set parameter by explicit value or by a deterministic expansion of an author-written pattern; put/delete definitions; wrap/unwrap). Proposals are validated against their exact base hash, executed on sample inputs and, for pure refactors, tested for exact behavioural equivalence. Makers may not run code that computes weights from data, may not script search loops, and receive development feedback only through a budgeted `dev-eval` command. A manual-maker operator records development train-pair diagnostics, authored transactions, actor/model provenance, graph hashes and candidate status. Each serializable proposal consumes a budget unit, including invalid, rejected and unchanged edits; transactions are applied to a copy and checked against graph validation, structural profiles and typed runtime fixtures. Three focused tests pass. The operator calls no LLM service and performs no benchmark itself, so a matched non-LLM mutation arm must use the same proposal budget and access. A frozen candidate receives one ledger-accounted final call, consumed even on failure. This is accounting infrastructure, not capability.

3.3 Breeding and evolution

Breeding uses checked `import_def` crossover: a child imports definitions from parents with namespace, shape, ownership and resource validation. The proposed and later implemented genome uses fixed slots with typed ports, bounded parameter shapes and shared parameter identities; a slot may exchange only a compatible definition, and crossover preserves dependency order, declared state and the two-attempt grid interface. Broken wiring should be prevented by construction; static validation remains a final safeguard and does not show task competence. Children commit parent hashes, seeds and donor-per-slot choices, and neutral lineage metadata does not change canonical genotype identity. Primary selection is discrete (μ, λ) population selection with Gaussian coordinate or block mutation; it is not a fitness-weighted gradient estimate. The owner later authorized automated numeric fitness selection and count-fitted seeds (dev-training corpus only), and then prohibited gradients of every kind and excluded the Hedge-style adaptive mixer from all primary

arms. Evolved parameters are therefore reported as fitness-selected rather than hand-authored, with donor provenance preserved.

3.4 Governance and evaluation boundary

A hash-chained ledger, graph version store and receipts make each run auditable; reported ledger-entry counts were verified after retrieval. Final cases for the digit task exclude, for strings of length ≥ 4, every string authors ever saw; the unseen-font split uses a font file never present on the authoring machine. Reserved audit data must never be shown to a maker, and schema guards alone cannot prove information isolation.

4 Experiments

Unless stated otherwise each experiment is a single run per arm.

4.1 Digit strings

Protocol. 32×128 grayscale, 1–8 digits, DejaVu Sans Mono 20 px, horizontal offset 2–14 px, vertical jitter ±2 px, Gaussian noise $\sigma \leq 0.03$. Final splits: iid ($n = 2000$), unseen font Liberation Mono ($n = 1000$), size 18–22 px ($n = 1000$), length 9 ($n = 500$). From an EOS-constant bootstrap: two independent iterative makers with recursive bundles (25 dev-eval budget each), a flat-graph maker, and a one-shot maker (validation only, no metrics). Controls: a gradient-trained slot-MLP (Adam, 400 or 20,000 training images) and (1+1) random direct-edit hill climbing (25 or 20,000 evaluations), all scored through the same reference evaluator.

Artifact	iid	unseen font	size shift	length 9
Authored, iterative A	1.000	**0.738**	**0.723**	1.000
Authored, iterative B	1.000	0.382	0.379	0.000
Authored, one-shot	1.000	0.427	0.453	1.000
Authored, flat	1.000	0.407	0.570	1.000
Gradient slot-MLP, 400	0.057	0.054	0.050	0.000
Gradient slot-MLP, 20k	0.838	0.100	0.419	0.000*
Random edits, 20k evals	0.000	0.000	0.000	0.000

Table 1: Digit strings: exact-match rate on independent final cases (fresh seeds). Wilson 95% intervals are ±0.03 or narrower at these n; CER in the project log. *The slot-MLP has eight character slots by construction.

Results. Table 1. All four authored artifacts are perfect in distribution, using two or fewer development evaluations each, so the in-distribution task is saturated and does not discriminate arms. Every maker converged on the same mechanism family (vertical alignment, left-edge detection, a fixed 12 px character pitch, hand-typed 15×12 glyph templates), yet small decisions produced large robustness differences: artifact A's normalized agreement score and ±3 px search tolerance generalized best; artifact B hard-coded eight character cells and failed every 9-digit string. LLM authors can construct compact recognizers (about two thousand parameters for artifact A) without gradient fitting, and the one-shot author needed no feedback. The results do *not* show a difference between arms (one run per arm), robustness to real-world text, or superiority over gradient training: the control lacks translation invariance and a gradient-trained convolutional network would likely be

perfect in distribution. The templates came from *looking at* development images; we regard the author's visual perception as the channel through which learning happens and report it as such.

4.2 Synthetic-grammar hierarchical mixture-of-experts language models (`tinylm/1`)

The owner judged digit recognition too easy a test of the intended cell→cluster→expert design, so authors built character-level language models with routing and recurrent state doing real work.

Task and arms. Text comes from a private English-like grammar (12 nouns, 12 verbs, agreement that must survive relative clauses, adjectives, prepositions, adverbs, ", and" coordination) held only on the evaluation server. Authors see 2,000 sampled documents and 300 development documents, never the grammar. Vocabulary: 32 tokens; the model receives the last eight characters, the position and its own state. Scoring is bits per character (bpc) under teacher forcing on fresh documents, plus grammaticality of 200 free-running samples (temperature 0.8) judged by the private recognizer. Three authors used an enforced ladder profile (cells inside clusters, clusters inside experts, a router driving lazy top-1 dispatch among at least two experts), two an open profile. Controls fitted from data on the same 2,000 documents: uniform, n-gram counts and a gradient-trained MLP over the same eight-character window (Adam, 74k parameters).

Model	bpc	top-1	gramm.	gramm.+novel	experts	params
Ladder L1	0.6796	.810	.966	.788	6	278k
Ladder L2	**0.6776**	.809	**.997**	.884	60	1.69M
Ladder L3	0.6790	.809	.940	.804	4	898k
Open O1	0.6797	.809	.955	.839	–	1.31M
Open O2	0.6808	.809	.939	.775	–	1.56M
Gradient MLP (window 8)	0.824	.780	.201	.135	–	74k
Trigram counts	1.383	.617	0	0	–	66k
Uniform	5.000	0	0	0	–	32
Generator entropy: ≤ 0.6734 bpc (summed choice bits, upper bound)						

Table 2: Synthetic-grammar language models: independent final evaluation on 1,000 fresh documents (bootstrap 95% intervals on bpc about ± 0.002). "Gramm." is the fraction of complete generated sentences accepted by the private grammar; "+novel" also requires the sentence to be absent from the authors' training text.

Results. Table 2. All five authored models reach 0.678–0.681 bpc, within 0.004 bpc (under 1%) of the generator's entropy, with 94–99.7% grammatical generations, against 0.824 bpc and 20% for the gradient MLP. The authored models keep singular/plural state across nested clauses in a stack held in recurrent state; the fixed-window MLP cannot. The best model (L2) has 60 routed experts, one per grammatical situation, and only 3 of 60 sampled sentences violate the grammar. The task is exhausted: every author read roughly 40 documents, wrote the closed grammar down, hand-tallied rates and compiled it into tables and state machines. This shows the machinery (state, recursion tracking, dispatch among tens of experts) works end to end at 0.3–1.7M authored parameters. It does *not* show that the ladder helps: an open, unrouted model (O1) matched the ladder with three feedback runs against five to eleven, and ladder authors were required to use routing the task does

not need. Nor does it show that authoring scales to data whose generator cannot be read off a few dozen samples. Author-reported cost: 176–231k tokens per run.

4.3 Real-text mixture of experts, delegation and failure-informed mutation

The next step moved to real text, where no closed grammar exists. In a delegated real-text model (hash `82ebe5e5...`), three isolated mutations reused settings from earlier failed proposals. The first worsened development bpc from 2.4748 to 2.4826. A trigram-donor change scored 2.4746 bpc, effectively tied with its parent; an after-period-space specialist scored 2.4743 bpc. These comparisons used the 165-document development set. The specialist was accepted on a separate eight-expert branch; its reference-anchor bpc was 2.46152 versus 2.46236 for the seven-expert parent over 155,239 tokens, with identical 0.5220 top-1 accuracy. Both checks are development feedback and establish no final or frontier-benchmark gain; the main branch remains seven-expert.

A UTF-8 byte lift of the eight-expert text MoE retains authored weights and 225-dimensional recurrent state in one graph with a shared 259-token head. Its inherited experts still use only eight projected bytes. A 387-token ASCII compatibility probe preserves 208 correct predictions, with loss changing from 2.4798989 bits per character to 2.4803443 bits per byte. This establishes input plumbing, not language competence; three ordinary or Unicode prompts repeatedly emit "too" to the token limit (receipts: `runs/unified-byte`). A directly authored repetition-memory mutation reading a 64-token context has parent loss 2.480344 versus 2.512589 bits per byte on the same 387 bytes; both predict 208 tokens correctly with zero scoring failures, and both answer 0/4 author-visible novel key/value binding prompts exactly. The mutation was rejected for promotion (`runs/parallel-reasoning/DECISION.json`); this small development probe is not a held-out reasoning evaluation.

4.4 MNIST-1D direct authoring and crossover

MNIST-1D provides a procedural ten-class task on 40-sample signals. We used 4,000 development-training signals, a separate 1,000-signal development evaluation set and a 1,000-signal reference anchor; independent final evaluation used 2,000 newly generated signals per artifact with an operator-side secret seed. Makers were told not to inspect the generator source, a restriction that was procedural rather than technically enforced.

Round A and controls. Three independent makers started from the same zero-logit bootstrap under harness version 1, wrote numerical filters and classifier wiring without gradient fitting, and used respectively four, five and four development evaluations. Table 3 shows the strongest authored parent exceeding the logistic control but trailing both MLP controls (trained on the same split). Class-specific strengths differ substantially; three runs are insufficient to estimate a typical outcome.

Crossover. A fourth maker imported the independently authored A1 and A2 filter banks into a child graph and authored a combining rule. After five development evaluations the accepted child scored 0.5295 on fresh final signals, an absolute gain of 0.064 over the better parent; it uses about 29.7 million multiply–accumulates over the final set versus 13.3 million for A1 and 16.4 million for A2. This is one selected offspring, not evidence of a general evolutionary advantage. A three-parent extension scored 0.572 on the development set and 0.557 on the reference anchor ($n = 1000$ each), then 0.5360 on a separate fresh-seed final set ($n = 2000$). Its apparent 0.0065 gain over the two-parent child is not reliable: the final evaluations used different seeds, the intervals overlap

Artifact	Final accuracy	Wilson 95% interval
Authored A1	0.4655	[0.4437, 0.4874]
Authored A2	0.4360	[0.4144, 0.4578]
Authored A3	0.3410	[0.3205, 0.3621]
Authored two-parent crossover	0.5295	[0.5076, 0.5513]
Authored three-parent crossover	0.5360	[0.5141, 0.5578]
Gradient logistic	0.3355	[0.3151, 0.3565]
Gradient MLP, 100 hidden units	0.5940	[0.5723, 0.6153]
Gradient MLP, 500 hidden units	0.5995	[0.5779, 0.6208]

Table 3: MNIST-1D independent final accuracy; $n = 2000$ per artifact. The MLP controls were not tuned to match published MNIST-1D baselines.

substantially, and the third parent raises final-set computation from 29.7 to 39.3 million multiply–accumulates. A version-2 diagnostic plugin reporting class confusion and wrong-prediction margins was validated and activated, but fresh-maker Round B under that harness was never evaluated, so no harness-effect claim is possible. New MNIST-1D jobs were paused under the later document-benchmark priority.

4.5 Frontier-benchmark probes at chance

We adapted character-level models to multiple-choice scoring (`mc-loglik/1`) by summing next-character negative log likelihood over each answer choice, also reporting length-normalized scoring. The vocabulary lowercases text and omits many capitals, digits and symbols, so coverage is imperfect. MMLU test, ARC-Challenge test and HellaSwag validation were staged on the private evaluation server; GPQA was unavailable because access requires dataset authorization.

The first sweep covers a synthetic-grammar authored model and three real-text controls (unigram, trigram, small gradient MLP), each on a fixed 1,000-item subset per benchmark. Length-normalized accuracies span 0.243–0.266 on MMLU, 0.224–0.248 on ARC-Challenge and 0.230–0.264 on HellaSwag. The delegated real-text MoE (`82ebe5e5...`) on 1,000 items each scores 0.241 MMLU, 0.243 ARC-Challenge and 0.241 HellaSwag (Wilson intervals $[0.2155, 0.2685]$, $[0.2174, 0.2705]$, $[0.2155, 0.2685]$; character coverage 0.9147, 0.9624, 0.9889; receipt `runs/bench_results/server-aggregate-20260930.jsonl`). All values sit around the four-choice chance rate of 0.25 and provide no evidence of frontier benchmark competence; they are negative transfer evidence and a test of the scoring interface, not a comparison to frontier language models.

4.6 Byte-level language graph, component specialists and the document hold

Tool-loop probes. The actual byte-language graph (hash `f0b43dbe...`) was asked to generate tool-loop actions on three authored synthetic prompts (one policy step, 48 generated tokens per action). It produced three invalid JSON actions and zero tool calls or responses (`runs/parallel-tools/server-policy-receipt.json`). This language-policy failure is retained alongside the following component results, each a narrow authored algorithm evaluated by the same research team and not evidence of language understanding:

- Typed finite-state graph (`7f810c68...`), `typed-observation-transfer/1`, 128 fresh numeric payloads of width seven (seed 90218, 640 policy evaluations): transfers 128/128 observations versus 0/128 with observations removed, abstains on 128/128 no-request and 128/128 invalid-

observation cases and calls on 128/128 missing-observation cases (`runs/parallel-tools/server-typed-receipt.json`).

- Exact binding graph (`e36653df...`, zero parameters): 128/128 server-generated lexical-variation cases (Unicode, missing keys, prefix distractors, duplicates) match, versus 0/128 for a disabled "unavailable" component (a dependency ablation, not a baseline); receipts in `runs/parallel-reasoning/binding`.
- Carry-addition graph (`43c2a56a...`): 63/63 development additions versus 2/63 for an identity control, all five invalid inputs rejected (`runs/parallel-arithmetic/REPORT.json`); its empty source-hash field limits provenance.
- Frozen digit OCR graph (`54e3934a...`) returns correct text for five authored digit strings; the negative control "hello" yields "50115" and blank input yields empty text (`runs/parallel-vision/server-smoke.json`).
- Positive-Horn deduction graph (`3e9019e4...`): 120/120 development cases (71 entailed, 49 not entailed) versus 56/120 with rules removed, 120/120 under rule-order permutation (`runs/parallel-deduction/REPORT.json`).

Composition. The action-grammar graph `8868695d...` answers two fresh synthetic digit images ("28", "39") correctly through the actual specialist harness, each with one OCR call and zero invalid actions; with one observation withheld it returns no answer and repeats the call twice, and with the tool disabled it returns no answer with zero accepted calls and two invalid actions (`runs/action-language/harness-development-result.json`). The composed graph `e04777f4...` preserves inherited MoE dispatch and gives the grammar graph-owned priority; it is bit-identical to its parent on one ordinary-input test, and generates OCR text "587" under preservation probes (four focused tests), and succeeds on two further digit images ("88", "86"). A matched 387-byte probe gives parent and composed graphs identical loss (2.480344 bits per byte) and 208 correct tokens. The source-stability flag is false because a batch method was appended during execution; the coordinator reports identical serial-method ASTs and the executed snapshot is retained in `runs/action-language/composed-runtime-snapshot`. A tiled variant `ab3cee4e...` preserving unused inherited arrays passes the strict micro profile with identical 387-byte scores; it retains 81,432 parameter slots, 3,868 nonzero values, 96 cells, 26 bundles, 613 instances and one dispatch, edges rise from 14,327 to 15,618, and a single local run takes 12.097 s versus 9.746 s untiled (about 24% overhead, uncertain). The controlled-English deduction composition `bef61dc6...` calls component `3e9019e4...` in an immutable five-condition harness (`runs/action-deduction/harness-snapshot-v1/result.json`): positive, missing-fact and missing-rule cases answer correctly (3/3), each with one call, zero invalid actions and zero specialist errors; withheld observations give no answer and two repeated calls; disabled tools give no answer, zero accepted calls and two invalid actions. Condition latencies are 163.1, 150.6, 144.1, 140.8 and 107.6 seconds. The grammar is bounded unary controlled English, and repeated unavailable calls and high latency remain failures. The coordinator reports 153 full-suite tests passing in 81 seconds; that is regression evidence, not capability.

Document benchmark hold. The priority benchmark is MMLongBench-Doc (repository); this is a research priority, not a claim that it is the hardest benchmark. All five OCR graph families are digits-only, fixed 32×128 raster interfaces with no question conditioning or layout encoding and cannot yet meet its contract; a digit-only proxy would establish nothing. The pinned source has 135 PDFs and 1,082 annotated questions (the README records 1,091; see Section 8). Private staging

completed with zero benchmark evaluations. The first split was rejected after two byte-identical PDF groups and one cross-split conflict (`runs/document-crossover/server-stage-v2.json`); canonical grouping gives 133 unique byte groups, provisionally 107 development and 28 held-out document IDs. The related-edition audit reports 67 candidate pairs, nine crossing the split, and 40/133 groups with inadequate extracted text; a later audit groups 117 heuristic families, which does not verify isolation. The split is not frozen. Alphabet-expanded OCR (`8332b0c6...`) recognizes three of six tiny development strings; its 61 literal-glyph checks test the templates, not accuracy. Three OCR crossover children are frozen without selection. The same-item diagnostic (`document-phase-diagnostic/3`) renders in 1.6596 s, completes all 2,125 OCR crops in 133 batches (1.7213 s), then enters language generation at 2.1396 s with a 783,438-byte context and times out at its 120-second limit (`runs/document-crossover/server-phase-diagnostic-v3.json`; the recorded `language_seconds=0` is an unfinished timer). A compact serialization (v4) cuts the context to 185,055 bytes (a 76.4% reduction) yet still times out after 1.7736 s of OCR (`runs/document-crossover/server-compact-diagnostic-v4.json`); this negative result selects no model. A compiled-runtime backend initially shows about 1.4× step speed; differential qualification (29 tests, 74-fixture injection suite; first NaN failure retained and repaired) reports amortized speedups of about 21–26× and first-request speedups of about 5–7.5× with matching states, logits, tokens and statistics on tested cases (`runs/compiled-runtime`); it is opt-in and is an implementation measurement, not capability. No document score exists. A dated comparator plan (`docs/FRONTIER_COMPARATOR_PLAN.md`) names endpoints `gpt-6-astra` and `claude-fable-5-1` (verified from vendor documentation on 29 September 2026); it matches ordered 72-DPI page rasters and questions, no paid requests have been made and endpoint access is unverified. The measured language failures motivated `docs/LANGUAGE_CORE_OPTIONS.md` (a trained shared trunk, the no-gradient primary, or a three-size study); the owner rejected gradient descent and the trained-core option, so the option is historical. Document scoring is held until a predeclared plain-text QA gate; a proposed 16/20 coherent-answer threshold is not frozen and is not a goal-completion criterion. General language capability, not transport, is the blocking question.

4.7 Evolution pilots on the byte MoE and count-fitted seeds

Generation-zero calibration. `runs/evolution-language-gen0-v2.json` records the founder `f0b43dbe...` on 334 content bytes at 2.042730829833363 bits per byte, zero failures, 9.81954 s single-thread nice-15 (about 34 content bytes/s on this workload). This is selection fitness on public development-training material with EOS excluded; V1's requested 150–256-byte stories selected no cases, and V2 used 288–350 after a 288–600-byte inventory, without using model outputs. A local founder curriculum receipt (`runs/language-curriculum/FOUNDER_DECISION.json`) on eight authored cases scores its 190 content bytes at 4.0663511910499714 bits per byte; raw copy, recall and QA each fail all two cases and all six generations repeat "too" to the 24-token cap (9.64137 s). These cases are separate from the 334-byte calibration, so the losses are not comparable.

Mini-pilot (null). `runs/language-evolution-minipilot-v1-launch.json` records a single-thread nice-15 run with a 2-GiB address-space limit and 600-second timeout: three generations, each proposing one router-array and one expert-array candidate, giving six GA evaluations and six founder-random controls on paired seeds; each proposal mutates eight nonzero coordinates with relative Gaussian scale 0.05 and survives only on strict development improvement. It completes in 152.67957 s with peak RSS 68,136 KiB (about 66.54 MiB), twelve candidate evaluations, two unique report graphs and a sixteen-entry ledger (`runs/evolution-language-minipilot-v1/result.json`). Both arms score 2.042730829833363 selection bits/byte at generations zero and one

and 2.0422575408863954 at generations two and three, choosing the same winner `e4d1e5db...`; report-only loss is unchanged at 2.606790482283684 with zero failures. There is no evolutionary advantage or report improvement, report data are public development material, and seven focused tests pass according to the coordinator. Twelve candidates over 152.68 s do not establish a pure candidate rate. Context/state mutations, zero-coordinate activation and crossover are proposed v2 work.

Count-fitted references and seeds. Side research (`notes/CLAUDE_RESEARCH.md`, `scripts/claude_count_reference.py`) reports byte-level order6 Witten-Bell loss 1.3816 on 165 development documents (89,213 bytes), while founder loss 2.5113 uses only 24 documents (12,894 bytes); the scopes are unmatched and establish no matched gap (Table 4). A feature script fitting on development-training documents except the final 150 (used to select mixture weights) reports 1.3909 count-only and 1.2650 with word context and document cache; a side adaptive mixer reports 1.2222. The mixer is a Hedge/exponentiated-gradient-style update, now prohibited and excluded from all primary arms. These are exposed development numbers with non-gradient grid selection, not QA or LLM capability. Server parity of the exact order6 Witten-Bell IR export on the first 24 development documents (12,894 bytes) gives IR and reference both 1.3852251087 bits/byte, maximum position difference about 2.27×10^{-13}; the fitted corpus excludes prior report families, so this is not an exact-scope comparison with 1.3816. Guarded cached scoring on 190 development bytes takes 0.473 s for a router mutant including certification/hash/compile and 0.302 s warm, with bit difference about 1.14×10^{-13}.

Scoring scope	bits/byte	Note
Order6 Witten–Bell, 165 dev docs (89,213 bytes)	1.3816	counts; side reference
Count-only, weights from last 150 train docs	1.3909	non-gradient grid
+ word context and document cache	1.2650	exposed dev feedback
Side adaptive mixer	1.2222	excluded from primary arms
Order6 IR vs reference, first 24 docs (12,894 bytes)	1.3852251087	parity check
Founder, 24 docs (12,894 bytes)	2.5113	unmatched scope
Founder, 334 bytes	2.042730829833363	server calibration
Founder, 190 bytes (curriculum)	4.0663511910499714	different cases

Table 4: Count-fitted references and founder losses. Scopes differ, so rows are not a matched comparison; all data are exposed development material.

Static count-seed four-arm smoke. The bounded static WB6 smoke run (two proposals per search arm) has zero acceptances and unchanged winners; fitted-evolution and fitted-random proposals duplicate at this budget. Eight cached/full qualification cases have maximum bit difference zero (39.986 s). The run takes 25.766 s for 24 batch evaluations (0.931 batch/s); six unique graphs including two baselines correspond to 0.233 unique graph/s end to end. Fitted corpus losses on batches A/B are 1.0819295/0.7147698 versus founder 2.271318/1.621583; fitted long-context fitness is worse than founder, and a 0.05 residual improves corpus loss slightly but worsens context and is rejected. Twelve ledger entries verify; CPU seconds and peak RSS were not captured (`runs/wb6-four-arm-static-v1/`). This two-proposal null result is not scaling evidence.

4.8 ARC-AGI-2 development: authored cell library, gene coverage and breeding ties

Protocol and exposure. Grid receipts use `arc-grid/1`: SHA256 task-ID hash order, with residue zero modulo five reserved for audit and nonzero for development. The scorer validates grids of 1–30 cells per dimension with colors 0–9, accepts at most two candidates per test input and computes exact task success (every test input correct within two attempts) with a Wilson interval. The exposed development manifest (DEV200) contains 200 public-training tasks with 222 test inputs; it is already development exposure, and relabeling a subset as audit does not make it untouched. Historical receipts state that the residue-zero audit was reserved, but later server access cannot be certified from local metadata. Two attempts must be produced by graph-owned selection, not host-side expert ranking. No official ARC-AGI-2 or tool-use score is reported, and the BFCL V4 agentic category needs a search backend not configured here.

First specialists. The `grid20/1` graph routes among 20 identity/transpose and cyclic-recoloring experts using demonstration reconstruction error; it and its fixed-identity control solve 0/200 (no invalid outputs, Wilson $[0, 0.0188]$). The `grid-associative/1` mutation combines eight dihedral experts with demonstration-inferred color associations in graph state and solves 3/200 (1.5%, Wilson $[0.0051, 0.0432]$) versus 0/200 with inferred recoloring disabled (receipts `runs/grid-associative/DECISION.json`, `runs/grid20/DECISION.json`). Fixed two-gene concatenation measures pair coverage but is not a top-two router.

Coverage and curriculum. The graph-gene operator measures 28 individual genes and three materialized pairs on DEV200; every pair solves 3/200, with no breeding gain. Pair throughput is about 51 tasks/s and 0.0195 CPU s/task, with whole-process high-water RSS 87,384,064 bytes. The dense curriculum has eight families with 200 selection and 100 validation tasks each (2400 distinct original fingerprints): authored selectors solve the five geometry/color families, adding the bbox gene solves crop, and both portfolios fail tiling and local fill. All four graphs solve the same 3/200 native tasks; only two native tasks receive a train-only color-family annotation (both solved), 198 are unclassified (one solved), so family transfer rates are unavailable. The selector is authored, not learned. Whole-process RSS is 168.3 MiB with 139.33 s measured scoring; the synthetic generators are CC0-1.0/MIT-0 and competition eligibility is unverified. The `grid-gene-evidence-v1` receipt records 300/300 synthetic successes each for authored fill and count genes but 0/200 native tasks each; the count gene has one demonstration-consistent native task whose query fails; portfolios stay 3/200. The largest-four component gene passes 100/100 synthetic but solves 1/200 native with zero fully consistent tasks (about 14 tasks/s, 140 MiB; `runs/largest4-evidence-v1/REPORT.md`). Dense synthetic success has not supplied native transfer.

Separate co-researcher library. A co-researcher's 57-cell host-enumerated depth-two library reports 53/1000 public training solves, 65 demo-consistent tasks and 12 false positives (18.4615% conditional on consistency, 1.2% of all training tasks), and a once-scored public evaluation result of 0/120 with no consistent chains. A later version progresses from 80/1000 training solves (v2) to 95/1000 (v3) with v3 evaluation still 0/120 and 12 false positives reported. These are read-only external receipts on exposed evaluation tasks, not a fresh final evaluation or CellForge graph result. Full training access also potentially exposes the modulo-five audit for a combined library despite the operator's zero audit calls. Source imports and v2 filenames do not pin the receipt-generating run; worker-row means are 1.52686 s training and 1.805416 s evaluation, not a verified 0.5 s CPU budget claim, and RSS is unavailable. Its across-query attempt criterion differs from per-input

acceptance, so counts are not a matched comparison. Prize-winning odds are assessed qualitatively as very low; realistic deliverables are a valid open entry and an honest paper.

Single-gene and composition measurements. Table 5 lists authored genes (synthetic denominators are original generated tasks; native means DEV200 unless noted). Each row is one frozen graph with verified ledgers and no gradients, public evaluation, audit or final access. The object-

Graph / gene	Synthetic	Native	Cost and note
Conditional state v2 (2 rank-cell params, 1002 state dims, 1000 adjacency rules)	300/300 vs 0/300 (updates off)	1/200 vs 0/200; none fully consistent	117 vs 115 tasks/s; 49.3 MiB; 12.03 s per run
Adjacency-then-fill serial child (195 slots, 1010 state)	300/300 vs 0/300 for either part	3/200 for all; 2 consistent, 0 false positives	46.88 vs 52.04 and 50.40 tasks/s; 30.97/27.26/28.17 s; 55.4 MiB; 57 slot nodes, 1650 expanded
Direction router (2 ranking params, 6 state entries, 207 edges)	200/200 selection, 100/100 validation vs 0	0/200; none consistent	142.17 vs 147.06 tasks/s; 55.48 vs 55.64 MiB
Object association child (+11 params)	300/300 vs 0/300	4/200 vs 3/200	58.37 to 9.82 tasks/s; 61.27 to 150.95 MiB
Same, 16-step variant (174 params)	300/300	4/200; 3 consistent, 0 consistent wrong	9.82 to 12.49 tasks/s; 0.10180 to 0.08006 CPU s/task; 150.95 to 109.32 MiB; 1564 vs 3004 edges, 205 vs 589 constants
Pixel scaling factor 2 / 3 (31 params each)	300/300 each	2/789 and 1/789 (expanded training)	403.23 / 393.26 tasks/s; 0.002481 / 0.002544 CPU s/task; 21.76 s; 76040 KiB
Graph-owned scaling selection (64 params)	300/300 vs 100/300 (updates off)	3/789 vs 0/789	182.42 tasks/s; 0.005483 CPU s/task; 47.72 s; 70896 KiB
Expanded active-slot graph (7 slots, 273 params, state 1515)	110/120 vs 70/120 (2 specialists active)	not scored	14.44 vs 44.31 tasks/s; 44.31 s wall; 103064 KiB; 40 paired gains, 0 losses
Homologous five-gene bank recombination (25 combos, 206 params)	child 10/10 vs 5/10 per complementary parent	not scored	14.12 tasks/s; 7.20 s; 113188 KiB

Table 5: Authored ARC genes and compositions on exposed development data. Native figures never exceed the 3/200–4/200 range on DEV200; the object gene's extra native task is exposed development only.

association gain was identified by a post-hoc frozen-model observer, complete paired regression attribution is absent, and the child adds operations without matched compute; the 16-step variant's paired check against the unchanged portfolio verifies one gained task and no losses across 200 tasks, and full pairing against 64 steps is unavailable (aggregate receipt). The two scaling genes' three disjoint successes were already identified from development demonstrations outside DEV200, so this is not new independent transfer. The direction router is an explicit maker-authored correction: a scalar lookup had matched the context graph on all 24 selection tasks with all 97 wrong pixels at unknown keys (94 border pixels, none in conflict with learned keys), and the authored router compares five hypotheses (identity and four one-pixel neighbor copies); authoring cost was not measured. A conditional-state v1 prototype without cells and the zero-cell generic version remain historical.

Training-only diagnostics and coverage census. On DEV200, demonstration-only diagnostics show associative-nine with 2/200 consistent tasks, 57 observed shape mismatches and 141

pixel-rule/routing failures; conditional state has zero consistent, 57 shape and 143 failures (17.014 s, 60.28 MiB, four ledger entries). These categories do not separate missing genes from inadequate routing. A structural census found 57 tasks with shape-changing demonstrations; a bank of identity, square tiling, pixel repetition at factors 2 and 3, and bounding crops explained no task exactly, and one color association added only two tasks the associative portfolio already reconstructed (0.35 s, 38.11 MiB). Extending the 17-rule census to 789 public-training tasks (nonzero SHA256 modulo 5; 211 residue-zero tasks excluded) gave seven exact matches and 13 with color association, two pixel-repetition tasks at factor 2 and one at factor 3 outside DEV200, no object-removal match, and no unique hole-filling match (5.68 s, 53596 KiB). These motivated the scaling genes; they are demonstration matches, not query solves. Derived cells must retain expanded-exposure provenance.

Matched population pilots. Table 6 collects every crossover, mutation and random-search comparison. Each is a single seed or a few seeds with four to sixteen proposals on exposed synthetic families, with winners frozen before reporting. Physical evaluation counts match across arms but CPU cost does not. No pilot shows a crossover-specific advantage. Details that change interpretation. In the object-bank pilot paired native receipts confirm one gained task and no losses versus either control, but crossover kept object and adjacency genes while random kept crop and adjacency, so equal totals conceal different niches. In the seed replication all 12 crossover proposals failed to combine object and adjacency, both founders died at generation 3 and two populations were extinct after generation 4; early founder loss is a hypothesis, not an explanation, and pure crossover can only reach alleles in living parents. The age-8 run exceeds the pilot horizon and does not support early founder death as the repair. In the seven-gene pilot all four crossover donor draws were $[0, 0]$ (copying one parent), so the complementary child was never evaluated; crossover and mutation share one winner and one report, so these are not independent. In the separate-streams run distinct winners retained repeat2/repeat3, crop/repeat2 and repeat2/fill, but stream, budget and seed changed together. In the seven-slot run founders scored 14 and 18 on selection, crossover first reached 22 at proposal 10 and mutation at 9, tiling failed in every arm and random also missed object tasks. In the context-projection pilot nine binary key-projection slots were varied with the center retained and all arms kept the orthogonal-cross founder; the graph stores 23 parameter slots. Population mortality and sharing were not separately ablated (shared graphs store 195 slots and 1010 state dimensions), the matched-v1 age/tie rules also changed, and the population-v2 policy lacks a native file-hash manifest, so external byte identity is not asserted.

Seven-slot capacity and breeding interfaces. A finite two-slot diagnosis evaluated all 49 layouts (24 selection tasks, 70 demonstrations, 3430 predictor calls; 285.20 s, 190632 KiB, 101 ledger entries): the demonstration-selected winner scored 14/24, post-hoc best single layout 14/24 and the query-oracle union 22/24 (failing both tiling examples); the union is not a deployable score, and one of two scaling examples per factor tests oversized-query identity fallback. These bounds motivated slot/wiring capacity rather than more search in a fixed space. The shared bank then grew to seven authored genes (49 two-slot combinations valid at 268 stored parameters and state 1010; homologous crossover preserves factor 2 and 3 from complementary parents; checks on 12 crossover and 12 mutation children; five tests), and crossover and one-slot uniform mutation operate on the expanded genome (16 reproducible crossover children and 16 mutations, tamper rejection, four children and eight ledger entries); no offspring fitness was measured for these interfaces. A bounded exact 3-by-3 context memory (23 parameters, 3602 state dimensions, four tests) is frozen but unbenchmarked.

Pilot	Synthetic report (crossover / mutation / random)	Native DEV200 and ledger	Cost
One-slot maker (`grid-maker-slot-v2`, seed 662021)	maker and control pick same fill gene and child; 600/800 to 700/800 (reconstruction 0.75 to 0.875)	3/200; 2400 demo-pair scorings per arm	172.02 s; 75,032 KiB
`grid-population-pilot-v2` (4 logical attempts)	140/180 each	3/200 each; 29 ledger entries; 9 unique genomes; cache hits 3/0/2	150.57 s; 73,804 KiB
`grid-population-matched-v1` (6 physical evals, 3120 calls, 18 evals total)	120/180, 140/180, 140/180	3/200 each; 36 ledger entries	228.63 s; 70,388 KiB
Context-projection (4 founders, 8 proposals, 24 tasks)	1/24 development, 0/24 report for all arms, 95.83% pixel accuracy	0 accepted proposals; 41 ledger entries	0.654–0.684 evals/s; 59.39 s; 123.54 MiB
Object bank (2 founders + 4 proposals, 348 calls per arm, 206 params, 1010 state)	14/20, 12/20, 14/20	4/200, 3/200, 3/200; 36 ledger entries	CPU 18.785, 9.704, 13.099 s; 11.59, 46.94, 47.15 tasks/s; 155.77 s; 140.66 MiB
Seed replication (3 seeds, 20 tasks, 18 evals, 1044 calls)	12/20 every run; mutation 14/20 throughout; random 14, 12, 14	not scored; 36-entry ledgers	CPU 38.675, 54.880, 45.086 s
Finite lifetime (max age 3 to 8, same seeds)	crossover 12/20; mutation 14,14,14 to 14,12,14; random 14,12,14	not scored	18 evals, 1044 calls
Seven-gene (seed 772081; 6 graphs, 420 calls)	12/24, 12/24, 14/24	not scored; 36 ledger entries	CPU 8.171, 8.262, 8.824 s; 0.734, 0.726, 0.680 evals/s; 30.34 s; 114324 KiB
Separate streams, 16 proposals (seed 772101; 18 evals, 1260 calls)	14/24 each (tie)	not scored; 108 ledger entries	CPU 26.575, 39.178, 30.947 s; 105.76 s; 152816 KiB
Seven-slot (seed 772201; 18 graphs, 1260 calls)	22/24, 22/24, 20/24	not scored; 108 ledger entries	CPU 89.416, 139.494, 118.018 s; 11.77, 11.70, 36.92 tasks/s; 372.55 s; 167512 KiB

Table 6: Breeding, mutation and random-search pilots on the ARC gene banks. Mutation and random figures follow the crossover column order. Ties dominate; the largest gap (22/24 vs 20/24) is a single-seed two-task difference.

4.9 Evolved cellular-automaton negative control

A co-researcher's scratch numpy experiments (not reproduced here; scripts are outside the repository) evolved tiny neural cellular automata with Gaussian mutation and discrete selection on ARC-AGI-2 public-training same-shape tasks, matched by random and identity controls. A 1488-parameter/T10 variant solved 0/40 tasks after 3000 evaluations per task; synthetic swap loss improved without a solve at 6000 evaluations. A 414-parameter/T3 variant fit synthetic training after 8000–14000 evaluations (1200 evaluations/s on a different six-by-six workload, not attributable to CellForge), with held-out success in one of three configurations, then scored 0/40 training and 0/40 test at 8000 evaluations per task with random and identity both zero; lower pixel loss is not task success. A 313-task ten-core run outside the two-core budget was initially reported pending, then corrected: it stopped at 30/313 with zero solves and is not a benchmark. Raw-weight evolution therefore solves no real task at these budgets, consistent with the need for good seeds noted in related work. The published gradient-trained NCA solves 23 of 172 feasible original-ARC tasks [6]; its GPU timing does not establish CPU performance or no-gradient feasibility. On CellForge hardware, a 1488-parameter/T2 graph passes 5/5 synthetic 30-by-30 identity probes at 93.23 adapter calls/s with 38,268,928-byte peak RSS, and a shallow residual graph (`c985a1ed...`, 414 parameters, 22 microcells, three iterations) satisfies the ladder-micro profile (eight combined and three focused tests) and returns 5/5 at 69.1217 two-phase calls/s with peak RSS 42,491,904 bytes (about 40.5 MiB), single thread and nice 15 (`runs/nca-shallow-v1/server-identity.json`). These are demo/query adapter calls, not ARC fitness; architecture, iteration count and dispatch differ, so parameter count does not explain the speed gap, and no ARC score, evolution or generalization is established.

4.10 Offline submission interface

An offline CPU submission adapter and source bundler have eight and three focused tests; bundling includes transitive dependencies and hash-manifest preflight. A server smoke test on two tasks and three queries takes 0.14686 s at 48,603,136-byte RSS and matches local JSON exactly (`runs/arc-cpu-package/README.md`). This establishes interface execution, not Kaggle eligibility or task accuracy.

5 Native-author baselines and literal-tensor studies, October 2026

The following studies have separate frozen protocols and denominators. We summarize preserved reports and saved comparison artifacts; manuscript preparation did not rescore their panels. "Native" identifies the platform author route, not a newly trained foundation model. The spatial primitives, task-family choice and composition topology were supplied by the host.

5.1 Negative ARC development result

The four-author spatial report records four complete replacements for the same historical parent in a fixed `ray_cast→tile_orient` chain. Authors received search-task demonstrations, previous tensors and one training failure, with no query outputs or development feedback. All 167 positions were required: 103 free values and 64 fixed orientation values. The same restrictions were frozen for native proposals and matched random tensors before authoring.

Two responses were invalid (a duplicate JSON key and a prohibited orientation value). One valid candidate scored 0/4 search and 0/4 development; the other scored 1/4 and 0/4. The hand-coded reference also scored 1/4 and 0/4; the historical parent, zero tensors and each of four matched

random draws scored 0/4 on both panels. Scored arms had zero runtime errors. Invalid responses were not scored or repaired, but consumed the full budget. Ranking used development successes, then search successes and cell ID, so this development panel was a selection panel, not an untouched final report. The search gain establishes no development transfer or advantage over the hand-coded reference. It is not an algorithm-invention, crossover or ARC qualification result.

5.2 Direct prediction on previously exposed ARC search tasks

The single-task direct-grid report and three-task direct-grid report record four independent, tools-free per-task Codex author calls. Each response directly specifies a query grid; it is not a tensor genome or a shared learned-weight model. The author receives the task's demonstrations and query input. Responses and task-bound receipts were frozen before trusted host label comparison, with no query feedback or retry.

Public-training SEARCH task	Direct exact	Pixel differences	Identity control
`bd283c4a`	1/1	0	0/1; 72 differing pixels
`1a07d186`	1/1	0	0/1; 18 differing pixels
`3194b014`	1/1	0	0/1; shape mismatch
`f5b8619d`	1/1	0	0/1; shape mismatch

Table 7: Direct-inference baseline on four previously exposed search tasks, one query and one author call per task. The 4/4 total describes only these selected cases; it is not an ARC benchmark aggregate or held-out estimate. Pixel differences are undefined for the identity control when output shapes differ.

All four direct predictions were exact with zero pixel differences and matching shapes (Table 7); the reported fresh-process replays matched every saved result. Identity is a weak control. A frozen host-inferred frequency-packing program also solves `bd283c4a`, and the prior literal ray/tile cell already solved `f5b8619d`. No advantage over a capable task-specific program is established. All four tasks have prior search/evaluation exposure, and public-training tasks may also be familiar from pretraining; the result does not establish new model knowledge, held-out transfer or benchmark completion.

The parent attests tools-free isolated authoring. Exact parent-relayed response text is preserved, but original platform-byte captures are unavailable. Reasoning is recorded as `xhigh`; the setting was the default model and its exact backend identity is unknown. Credits are unknown. The four direct-baseline calls are accounted separately, with no genome positions credited to the literal-weight ledger. The data-only scorer executes no authored source; reserved report and excluded evaluation data were not used. These results leave the existing weight-population qualification failures unchanged. The following newly selected training panel is reported separately because its exposure history differs.

5.3 Direct prediction on newly selected public-training tasks

The batch03 report records four additional exact direct predictions and four matching fresh-process replays. Within this selected panel, the split is ARC-AGI-1 2/2 and ARC-AGI-2 2/2 (Table 8); these are sample counts, not version-wide benchmark scores. There was no query-feedback revision, retry, selection from query outcomes or extra author call.

Selection seed 202610060303 was frozen before reading new task contents. The report's official training manifests contain 400 ARC-AGI-1 and 1,000 ARC-AGI-2 IDs, with 391 shared. A local

Training source	Task	Exact	Pixel differences	Identity control
ARC-AGI-1	`46442a0e`	1/1	0	0/1; shape mismatch
ARC-AGI-1	`c8cbb738`	1/1	0	0/1; shape mismatch
ARC-AGI-2	`984d8a3e`	1/1	0	0/1; 54 differing pixels
ARC-AGI-2	`0c786b71`	1/1	0	0/1; shape mismatch

Table 8: Newly selected official public-training sample, four separate direct author calls. Direct predictions have zero pixel differences and all saved-result replays match. Identity scores 0/4 and is only a weak baseline. Newly selected does not mean proven unseen.

text-record inventory excluded 238 and 644 known-record IDs respectively; shared IDs were also excluded from the ARC-AGI-2 pool. The remaining pools contained 162 and 203 IDs. Two per version were selected by the predeclared SHA256 order of `seed:dataset:task_id`, and selection was independently reproduced. Training files were pinned to ARC-AGI-1 commit `399030444e0a b0cc8b4e199870fb20b863846f34` and ARC-AGI-2 commit `f3283f727488ad98fe575ea6a5ac98 1e4a188e49`; both repository licenses are Apache-2.0. The report records pinned Git-blob matches on acquisition and SHA256 checks at scoring. No public/private evaluation or reserved report task was accessed.

These tasks were absent from the scanned local text records. Remote history, nontext history and model pretraining exposure remain unverified, so the panel is not a proven unseen or held-out evaluation. The parent reports new isolated tools-free author threads, each receiving the common instructions and its task packet without query labels. All used `xhigh` reasoning with the default model; exact backend identity, platform prompt bytes and credits remain unknown. Four parent-relayed raw responses and task-bound receipts were frozen before label scoring. No model-authored source was executed, and zero genome positions were counted.

Final direct batch and cancellation. The pinned batch04 report adds three exact submissions: ARC-AGI-1 `ed36ccf7` and `ef135b50`, and ARC-AGI-2 `712bf12e`. All three have zero direct pixel differences and matching reported fresh-process replays; identity solves 0/3. Seed 202610060404 was frozen before task contents, with prior selections and cross-version duplicates excluded using the existing official pins. The fourth planned task, `e7639916`, was cancelled during creation; the parent reports that no author was created. It is unscored, not an accuracy failure or success. Batch04 therefore reports 3/3 scored and 3/4 planned completed. Its STOP checkpoint allows zero additional author calls and no cancellation retry or automatic next batch. No query-feedback revision occurred. Model/backend, prompt-byte and credit limitations remain the same as for batch03.

Across the two exposure groups, the descriptive total is 11/11 unique public-training tasks from eleven independent per-task author calls, with 11/12 planned tasks completed. Four scored tasks were previously exposed search cases and seven were newly selected. This small, nonrepresentative sample does not establish a benchmark score, new model knowledge, shared weight learning or generalization. The calls produce task-specific grids, not a reusable neural-weight cell. The earlier exposed panel and the two newly selected batches retain separate reporting; none changes the negative literal-tensor development result or the unsuccessful population qualification.

5.4 Synthetic rule transfer: success within one supplied family

The separate synthetic transfer report uses an independently implemented oracle that fills occupied columns and assembles reflected tiles. Six demonstrations and twelve search cases used colors 2, 4

and 5 at dimensions 3–5. The 24-case development panel contained eight unseen-color cases, eight cases forming four color-permutation pairs, and eight larger rectangles at dimensions 6–9. Exact and recoloring/flip-equivalent layouts were excluded across panels, except the intentional pairs. The host deliberately chose an oracle family expressible by the supplied ray and tiling code.

Both founders scored 12/12 search and 24/24 development; the hand-coded reference matched them. Both matched random controls and zero tensors scored 0/12 and 0/24. All runtime-error counts were zero and all four permutation checks passed. Native and random winners were selected on search alone before one-time development opening. The prespecified criterion required 24/24, the permutation checks, and a margin of at least 6/24 over every matched random draw; it passed. Both founders were search-perfect, so neither of the two permitted failure-replacement slots was eligible. Authoring stopped after two completions.

The two numeric genomes are *functionally identical*: sixteen orientation maxima differ (9 versus 1), but every forward decision is the same, as documented in the saved behavior comparison. Full-column filling also makes vertical reflection redundant. These are two author completions with one behavior, not two distinct acquired skills. This is transfer to new palettes and sizes within one supplied synthetic rule family, with no mutation, crossover, advantage over the reference, or demonstrated ARC transfer.

5.5 Two-specialist composition with an identity router

The composition report inherits the selected 167-position column specialist unchanged and authors a 167-position horizontal specialist. The latter fills occupied rows and tiles them with vertical reflections. The two families use the same input distribution and output dimensions, so the demonstrations must identify the applicable family. After the horizontal founder scored 12/12 search, its ordered roster with the column specialist was frozen. A second founder supplied a six-position router over host-computed negative mean leave-one-out demonstration errors:

$$W = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad b = (0, 0).$$

Thus the authored router is literally the fixed identity baseline. It selects the expert favored by the existing demonstration-error features; this is not learned-router superiority.

Frozen composition arm	Exact tasks	Errors
Authored router and both native specialists	24/24	0
Fixed identity router / independent reference bank	24/24	0
Matched random router with native specialists	12/24	0
Column alone / horizontal alone	12/24	0
Zero router / constant expert choice	12/24	0
Zeroed / matched random horizontal specialist, identity router	12/24	0

Table 9: Fresh synthetic composition panel; each slash-separated condition has the stated score. Twenty-four tasks each contain three demonstrations and two queries. Task success requires both queries to be exact; the native bank made 48/48 exact query predictions.

Both stage founders were search-perfect and stopped without replacement. Search-only selection preceded final access. Table 9 shows a 12/24 margin over either single expert and the sole matched random router, exceeding the frozen 6/24 margins; all four permutation checks passed. The bank routed twelve tasks to each expert and scored 8/8 in each color, permutation and larger-rectangle

stratum. Its margin over identity is exactly zero (literal-router comparison). A perfect random router would have blocked comparative acceptance under the unchanged rule. One random draw does not estimate the typical random-router distribution. The result supports bounded composition of two complementary specialists, not sustained population evolution.

5.6 Provenance, controls, accounting and exposure

For the preceding October 6 studies, the reports link raw responses, charged attempts, source and protocol hashes, controls, selection records and fresh-process saved-runtime replay. Those replays matched the recorded results; this manuscript does not claim a new independent replay. Four initial spatial author threads failed at startup before producing candidates; the parent resumed those same threads as tools-free authors. Their common author01 context is parent-attested. The two synthetic-transfer founders were also tools-free with no inherited context according to the parent. Composition packet delivery was similarly attested. *Platform isolation and actual prompt bytes were not independently captured by the evaluator*; hashing supplied packets and responses cannot establish that boundary or exclude pretraining contamination.

The spatial and synthetic-transfer authors are recorded as `gpt-6-luna`, with reasoning effort unknown. The horizontal composition founder is recorded as `gpt-6-luna`/xhigh; the router's model and reasoning metadata remain unknown. The spatial report preserves an author02 host transcription error and its correction from the same supplied response before slot consumption. This was not a new author attempt or post-score repair. No other response was repaired or normalized.

Study	Completions	Positions/arm	Frozen maximum/arm
ARC spatial batch	4	668	668
Synthetic transfer	2	334	668
Two-specialist composition	2	173	346

Table 10: Native and matched random arms incur equal literal-position charges, including fixed, inactive and invalid positions. These are accounting units, not inference credits or matched CPU cost. Unused conditional replacement slots were ineligible, not silently discarded attempts.

Table 10 counts all submitted tensor positions before validation. Native and random specialist arms share the 103-free/64-fixed vocabulary; composition random routers draw all six coefficients uniformly from $[-9, 9]$. The exported bank contains 340 stored positions: 167 inherited column values, 167 horizontal values and six router values. Only the latter 173 are new composition charges. The prior native ARC sequence, including the spatial batch, totals 1,206 per arm; adding the separate synthetic transfer and composition categories gives 1,713 per arm, excluding belief trials and the separate October 8 NCA tranche below. The owner's stated ceiling was 62,000 Codex credits. *Exact credit consumption remains unknown*, so these counts do not demonstrate credit-cap enforcement, equal currency expenditure, or an efficiency advantage over external systems.

The synthetic protocols froze inputs, oracle/source, seeds, criteria and control vocabulary. Search winners were saved before opening the sealed panel or materializing its labels; runtime audit guards enforced that order. Query labels and family tags were stripped before bank prediction. This was a procedural seal, not encryption or an operating-system access barrier. Both development panels are now spent and cannot become untouched evidence for later tuning. All reported gains are conditional on these small, deliberately compatible synthetic families. There was no private ARC evaluation, new public-evaluation authorization, or population admission.

5.7 Fixed T8 literal-weight feasibility tranche

A separate bounded experiment on 8 October asked an LLM to supply every one of 1,488 literal scalar positions in one shared neural cellular automaton (NCA). Unlike the historical evolutionary control in Section 4.9, this study used a frozen eight-update architecture and at most two author invocations: one founder and one conditional, training-failure-informed mutation. The terminal report is `reports/NCA_TRANCHE_RESULT_20261008.md`; the preserved run is `runs/weights-nca-author-tranche-20261008/`. These new receipts are local artifacts accompanying this revision, not evidence claimed to exist in the previously cited repository commits.

Frozen architecture and exposed analysis population. The host supplied a 30-by-30 canvas with 12 float64 state channels, a fixed 3-by-3 neighborhood, a 12-unit ReLU hidden update and a linear output update. One set of 96 arrays is shared across locations, time steps and all cases. The decoder chooses the maximum among ten color channels after eight updates, with output dimensions fixed to the input. Authors could change only literal values from $\{-1, -0.5, -0.25, 0, 0.25, 0.5, 1\}$, with no task routing, demonstration conditioning at inference, gradient fitting or executable structure changes. This is a shared fixed-host weight experiment, not one separate network or program per task.

The exposed public-training inventory contained 11 tasks and 32 demonstrations. Five tasks, containing 15 demonstrations, require output-size changes that this decoder cannot produce. Before authoring, the protocol therefore froze an explicitly restricted analysis of the remaining six tasks and 17 demonstrations, while retaining the full 11/32 inventory accounting. Shape compatibility is a necessary condition, not proof that the remaining transformations are expressible. No query outcomes were scored. Here an exact task means that all its included training demonstrations are exact; it is not the query-based task metric used in the direct-prediction studies. These exposed demonstrations supply development evidence only, with no untouched-test or broad ARC performance estimate.

Founder and completed comparisons. The founder passed literal validation and exactly duplicated the fixed identity tensor, both numerically and in every prediction. Table 11 reports all completed arms. Mean normalized pixel error is the unweighted mean of each demonstration's fraction of incorrect pixels; lower pixel error does not imply an exact solve. The founder's unrounded error was 0.2877535467628347, exactly the identity reference value. Its saved independent fresh-process replay matched every prediction, summary metric, graph hash and injected-tensor hash. The first paired random tensor and its completed result were reused unchanged from the frozen preflight, without a redraw or re-evaluation.

Completed arm	Exact demos	Exact tasks	Pixel error
Authored founder / fixed identity	0/17	0/6	0.28775
Fixed neighbor smoothing	0/17	0/6	0.60229
Zero tensors	0/17	0/6	0.54658
First paired random tensor	0/17	0/6	0.95265

Table 11: Fixed-T8 analysis on exposed training demonstrations. Every completed arm was valid and had zero runtime errors. Each slash-separated condition has the stated score. The timed-out mutation and uncomputed second random arm have no accuracy measurement and are excluded from this table, not assigned zero scores.

The predeclared improvement criterion required strictly more exact demonstrations than the best fixed reference and every charged random comparison, with the selected reference's solved demonstrations preserved. A mutation also had to improve on the founder while retaining its solved demonstrations. The completed founder comparison did not satisfy the criterion; the identity duplicate remains in the record and incurs its full charge.

Conditional mutation and terminal timeout. The only mutation received the unchanged founder context, all 1,488 parent literal positions and the 17 ordered training failures. Its frozen 27,507-byte packet contained no query answers or fixed-reference/random outcomes. The invocation reached the predeclared 120-second wall limit: its immutable capture records 120.056 seconds, exit -15, both output streams closed and empty stderr. Only start events were captured, with no final response or usage event. Read-only recovery obtained those receipts after a transport reconnection; the invocation was not relaunched and no owned process remained. No response was fabricated, repaired, parsed into a candidate or scored. Thus there is no mutation tensor, accuracy measurement or observed mutation gain.

Both scientific CLI invocations consumed their full position budgets, totaling 2,976 positions in each proposal arm. Four prior non-scientific diagnostic CLI invocations remain separate overhead. The second paired random arm incurred 1,488 positions but was never computed because the terminal halt prevented further evaluation. *The two-attempt numerical paired comparison is incomplete*; equal charged positions do not establish equal executed control work, backend request counts, latency or monetary expenditure. Backend-internal request counts and monetary costs are unknown. The unchanged 15% subscription usage display is too coarse to infer a cost. The closed tranche permits no additional author invocation or retry.

Audit boundary and interpretation. The complete observed founder event stream passed the zero-tool audit; the incomplete mutation stream cannot establish accepted zero-tool completion. Preventive tool isolation was not established, and generic platform context and built-in skill descriptions remained. The frozen founder packet also had an implicit optimization objective and prohibited "fitting" without qualifying that term as gradient fitting. This prompt-design weakness was recognized after the founder result and recorded without retroactively altering either packet. Neither the ambiguity nor the timeout is a demonstrated explanation for the founder's identity output. The result provides no positive literal-weight capability evidence, and it does not prove architectural incapacity.

Separate locality lower bound. A subsequent read-only diagnostic, recorded in `reports/NCA_EQUIVARIANCE_DIAGNOSTIC_20261008.md` and `reports/NCA_LOCALITY_LOWER_BOUNDS_20261008.json`, groups the 2,061 included output pixels by identical radius-T input-color and mask neighborhoods. For the frozen shared deterministic 3-by-3 update, equal neighborhoods imply equal states after T steps and equal pointwise decoded colors. Padding remains distinct from in-grid color zero, and no coordinates, global operations or nonzero auxiliary initialization supply additional information. In each class, at least its size minus its largest target-color count must be wrong, even for an arbitrary local function. That function class contains the permitted NCA functions, so its minimum error is a lower bound on NCA minimum error.

The resulting unavoidable-error lower bounds at radii 0, 1, 2 and 3 are 372, 112, 7 and 2 pixels, respectively, out of 2,061. Two radius-three contradictions occur within task `1a07d186`: identical local inputs require different output colors. Thus at least four updates are necessary to reproduce this panel exactly under the stated assumptions. These are pooled pixel counts, not

the per-demonstration mean errors in Table 11. Radius four is the first with no contradictory classes; this does not prove sufficiency. At radii seven and eight all signatures are unique, making label consistency vacuous. Independent window-based reproduction confirmed the counts and the two radius-three witnesses, with six focused fixture tests passing. This mathematical diagnostic neither establishes T8 capacity nor explains the identity founder, and it supplies no new authored candidate or task score.

5.8 Prompt-contract comparison remains inconclusive

A separately frozen two-founder study tested a narrower design question: whether explicitly stating the objective and clarifying permitted numeric reasoning changed authoring under the same fixed-T8 interface. As documented in `reports/NCA_PROMPT_COMPARISON_RESULT_20261008.md`, A received the original 16,752-byte founder packet. B retained its architecture, tensor schema and 17 exposed training demonstrations, adding only an explicit exact-demonstration objective with a pixel-error tie-break and clarification permitting manual numeric reasoning while forbidding tools, gradients and runtime fitting. Both used the same requested `gpt-6.1-sol`/low route, 120-second wall limit, 2 GiB memory limit, seven-value vocabulary and 1,488-position budget. Order was fixed A then B, with no cross-arm or control feedback. The original 11-task, 32-demonstration inventory and six-task, 17-demonstration analysis restriction were unchanged; no query outcomes were scored.

A was a fresh invocation with separate capture and inference receipts, not a reuse of the historical founder as another observation. Its final response nevertheless matched the earlier 4,597-byte response exactly. The new tensor again equaled identity numerically and on all panel predictions: 0/17 exact demonstrations, 0/6 exact tasks and mean normalized pixel error 0.28775. The newly drawn matched random seed 2026100803 also scored 0/17 and 0/6, with error 0.87856; both executions had zero runtime errors. Fresh-process replay matched all outputs, metrics and tensor hashes and independently confirmed the specified random draw. Fixed identity, smoothing and zero receipts were reused unchanged. A again failed the strict feasibility-improvement criterion.

B timed out at 120.118 seconds without a final response or usage event. *B has no numeric score, and the B-minus-A effect is undefined.* The terminal stop prevented construction and evaluation of its paired random seed 2026100804. Both invocations retained their full charges, totaling 2,976 positions per proposal arm, without a completed paired numerical comparison. The two NCA studies together comprise four scientific invocations; the four earlier diagnostic invocations remain separate overhead. Backend request counts and monetary cost are unknown. No invocation was relaunched and no replacement candidate or further call was introduced after the stop.

A's complete accepted trace supports observed zero tool use; B's partial trace does not establish accepted zero-tool completion or absence of internal generation. As in the first study, tool prevention was not established. The timeout cannot distinguish planning, generation latency, provider delay or literal-output burden, nor show that the clarified prompt helped or harmed. Even with two completed arms, one fixed-order pair on exposed training examples would provide descriptive evidence only. With B missing, the comparison is inconclusive and supplies no causal prompt explanation, T8 capacity conclusion or positive shared-weight result. The prior tranche remains closed and unchanged.

5.9 Operational dispatcher checks are a separate result

At composition evaluation time, `specialist_runtime` still rejected `ray_cast`. The later bounded ray-adapter report documents a separately reviewed local admission while preserving the exact exported bank, template code and evaluator-supplied override tensors. It bounds ray launches by

$8hw$ and path steps by $8hw(\max(h, w) - 1)$ for each $h \times w$ input, including recursively staged demonstrations and all possible routed experts. Combined grid visits, launches and path steps must fit 1,000,000 work units; the eight-expert, depth-two, 30-by-30 intermediate and 4,096-leaf-call limits remain. These conservative work units are not FLOPs or currency. Deadlines are cooperative checks, not interruption of an executing numerical function.

Seven focused tests passed in 0.087 seconds on separate fixtures, including two new 2-by-3 column/horizontal cases, override equivalence and rejection before prediction when budget is insufficient. The adapter did not rescore the 24-task panel, establish new transfer, bind a default service, or deploy an agent. Large inputs that fit the experimental scorer can still fail this adapter's conservative limits. Operational admission does not retroactively validate a scientific claim.

5.10 Qualification remains unsuccessful

The reports preserve successful focused tests and unsuccessful broad qualification. During the initial 6 October manuscript pass the unmodified required command was attempted with one numerical thread and a 30-second child limit:

```
python -B -m unittest tests.test_one_standard tests.test_standard_semantic -v
STATUS: TIMEOUT
ELAPSED_SECONDS: 30.000
```

The log records seven contract/milestone failures and semantic fixture errors before timeout (complete manuscript-pass output). Earlier diagnostics record seven failures in fifteen contract/milestone tests and a semantic temporary-directory access denial; the timeout is not a completed semantic check. Missing evidence includes sufficient seeds and founders, reconciled full-study proposal budgets, increasing accuracy across generations, crossover children, mutation-over-random advantage and 100 distinct genomes. These narrow studies do not remedy those failures. Tests, floors and security settings were not changed. A historical server ARC-goal pass in `LAB.md` pertains to its own saved public-training package and does not qualify these experiments or establish a single model across ARC-AGI-1/2/3. The later direct-baseline documentation update did not rerun these gates or any scientific panel.

Newer cloud qualification receipt, 8 October. The published integrity report records a completed rerun after retrieving all 19 unchanged tracked standard manifests (16 top-level manifests selected by the gates): 25 tests, seven failures and three errors in 1.681 seconds. The failures concern budget reconciliation, seed count, improvement across generations, crossover, founder/niche coverage, mutation-over-random advantage and genome diversity. The semantic errors are `KeyError: 'evaluator'` in incomplete frozen manifests with empty founder/genome lists. They are distinct from the earlier desktop permission errors and timeout. No manifests or qualification tests were changed; the full standard remains red.

Terminal NCA qualification receipt. The completed tranche's later logs under `reports/nca-tranche-terminal-20261008/` record 72 passing focused integrity tests in 0.295 seconds and a fresh required qualification run of 25 tests in 1.796 seconds, with the same seven failures and three evaluator-key errors. The focused passes establish checks on intake, event handling and feasibility machinery; they do not qualify the scientific population or demonstrate a capability gain. No protected qualification test or floor was changed.

The accompanying `reports/QUALIFICATION_FAILURE_TAXONOMY_20261008.md` distinguishes declaration defects from scientific milestones. At that diagnosis, the partial workspace contained

18 discovered declarations omitting the evaluator key and containing empty population arrays. The three semantic exceptions repeat the same missing-key access in three checking methods; they are not three independent prediction failures, and later declarations are not exhaustively checked by those methods. The seven failures combine budget/seed declaration shortfalls with missing registered evidence of improving generations, crossover, founder coverage, mutation advantage and genome diversity. Empty registered populations do not mean that no tensors or author-call receipts exist elsewhere. Supplying a placeholder evaluator or altering protected tests would not establish the missing scientific evidence. The separate prompt-comparison closure records 120 passing focused tests in 4.173 seconds under `reports/nca-prompt-comparison-result-20261008/focused-tests.log`. The full qualification command was not rerun for that closure: its latest separate result remains 25 tests, seven failures and three errors.

5.11 Control execution must remain distinguishable from task failure

Read-only inspection of source at commit `02beaf18` identified two distinct limitations. In `structcode.evaluate`, the zero and random control calls retain fit counts but discard their error lists. An execution error can consequently appear as a low-scoring control and support the `controls_ok` flag. This is a prospective acceptance defect: weight dependence requires valid, correctly shaped counterfactual execution, not failure to execute. The capability checkpoint already requires preserving these errors before new proposals.

Separately, the pinned `standard_checks._zeros_like` recurses through lists but maps a dictionary to scalar zero. `check_weights_matter` passes the complete weights object to that helper, and its `solved` helper converts evaluator exceptions to false. For dictionary-valued weights this can conflate a malformed ablation with a valid zero-weight loss of accuracy. A historical ARC-goal pass alone therefore cannot establish valid zero-control execution.

These source findings neither recompute nor revoke every historical score. Their impact on a particular result requires replay of its exact saved evaluator and tensors with structure-preserving controls and explicit error outcomes. Original receipts must remain unchanged. Repairs and focused regression checks are implementation evidence, not new ARC scores or retrospective scientific qualification; protected gate changes require separate owner approval. This manuscript pass did not modify those gates or execute historical model-authored programs.

Prospective acceptance correction. Commit `bfc928bf` now preserves zero/random control fit counts and errors, retains random seed attribution, and requires error-free execution of all four control arms before comparative acceptance. Pilot feedback distinguishes control execution errors from valid control successes. The verification report records 28 focused tests passing, including real-sandbox invalid and valid controls. The owner specifically approved correcting the raw-LUT regression expectation: its invalid random outputs must cause rejection. A separate total modulo-10 fixture checks legitimate acceptance. No qualification test was changed, and the separate protected dictionary-ablation defect remains unresolved. This correction establishes prospective control handling, not a replay of historical scientific results or new capability.

Search-only headroom and an unresolved source binding. The 8 October `reports/ROUTER_HEADROOM_DIAGNOSTIC_20261008.md` audit computes identity and expert-oracle success as 12/12 from the previously exposed router author packet; each expert accounts for 6/12. Its nine distinct exact feature vectors contain no contradictory success targets. Thus the recorded search feedback offers no observed improvement headroom over identity. This is packet arithmetic, not independent expert execution or a new final-panel result.

Attempted independent search replay stopped before expert execution because the frozen protocol requires `skillcore.py` SHA256 `3f6b926ecb7650f041c2e0d8ee42dc0271588c76701ce9e30414b3007bcc5b8c`, whereas the pinned repository copy hashes to `ec3b064c1022a7eb4e97afa4faeadf313c555c9e722266ecb0868fce5fc0b40a`. The protocol is preserved at the same commit under `HOST_EVALUATION/PROTOCOL.json` in that run. A bounded relevant-path history lookup and newline-normalization check did not recover the required bytes. The cause is unestablished. Historical reports of matching replay remain attributed to those reports; independent reproduction from this imported snapshot is not established. No substitute runtime was executed and no historical hash was rewritten. Recovering the original source is a prerequisite for a new tensor-replay claim.

6 Discussion

What worked. Direct authoring produced working recognizers and compact language machinery without gradients: digit strings read perfectly in distribution, with one artifact retaining 73.8% under an unseen font, and authored synthetic-grammar models reached within 0.004 bpc of the generator entropy. Typed crossover retained parental building blocks on tiny probes and lifted MNIST-1D accuracy by 0.064 in one selected child. Narrow authored components (typed transfer, binding, deduction, addition, action grammar) execute exactly within their grammars, and a compiled backend gives large verified speedups. The later native studies add synthetic transfer within a supplied spatial family and composition of two specialists. Matching the hand-coded reference and fixed identity router sharply limits what those positive results explain. The separate direct-prediction baselines solve four exposed search cases and seven newly selected public-training cases. Their descriptive 11/11 scored (12 planned, one cancelled) demonstrates per-task inference success on a small, nonrepresentative sample, without proving unseen-task transfer or a reusable learned-weight artifact.

What did not. Authored and control character models sit at chance on public multiple-choice probes; the byte language graph emits invalid actions and repeats tokens; document evaluation is blocked on language capability and on an unresolved related-edition overlap; evolution from authored founders tied random search in the language mini-pilot and in most ARC pilots; dense synthetic success (300/300) did not transfer to DEV200, where every portfolio remains at 3/200–4/200; and evolved NCAs solve 0/40. The gradient MLP beat the best MNIST-1D authored model. The later four-author spatial batch again has zero ARC development gain. The fixed-T8 founder is an identity duplicate with zero exact demonstrations; its sole mutation times out, leaving the second random comparison uncomputed. A fresh original-prompt founder also duplicates identity, while its explicit-objective counterpart times out. That prompt comparison is inconclusive; these studies add no positive shared-weight result and do not resolve representational capacity. Neither successful synthetic batch required a mutation, neither demonstrated crossover, and no sustained evolutionary improvement follows from them.

Honest limits. Independent server final evaluations in the early experiments exist only for digits, synthetic grammar and MNIST-1D; their ARC, byte-language and document results are development results on exposed public data. The later synthetic final panels were local and procedurally sealed, with author isolation parent-attested, and do not establish ARC accuracy. Runs are single small runs (one per arm in most cases), confidence intervals are wide, and the two-task gaps in Table 6 are not evidence. Authors and orchestrator share an operating-system account locally, and

the orchestrator has administrative access to the evaluation server, so the final-evaluation boundary is physical rather than adversarially audited. Token usage is self-reported; the digit task used a single font family in development; the visual-retrieval memory arm and the MNIST-1D harness-version comparison were never run; the MNIST-1D crossover has one independently tested child and no random-crossover control, so its advantage may reflect selection and extra computation; a co-researcher's library results and scratch NCA numbers are not reproduced by us. The authored models transfer competence from the author LLM's pretraining and must be reported as such. There is no private-set score, no official ARC-AGI-2 or BFCL score, the system is not an LLM, and no claim of general intelligence or frontier superiority is made.

Design-only pretrained composition. A System 1/System 2 proposal (`docs/SYSTEM12_DESIGN.md`) would compose pretrained models (a gradient-trained 0.6B-parameter decision-head system and a pretrained generative candidate) with evolved routing. It would import pretrained capability, is separate from the no-gradient line, and was not executed: no model was downloaded, and the approximately 4.34GB required files exceed the reported 3.5GB free disk.

7 Roadmap beyond the current results

Next ARC-AGI-1/2/3 agenda (proposed, not executed here). For ARC-AGI-1/2, retain a task-ID-deduplicated public-training manifest and explicit exposure history; freeze search, selection and untouched reporting roles before proposing reusable cells. Test whether synthetic gains transfer to new authorized task families against zero tensors, matched random attempts, fixed identity routing and the same hand-coded reference. Report task accuracy and failures with original denominators, rather than combining historical split scores. For ARC-AGI-3, the documented shared-policy probe solved zero levels on three public development games at 1,500 actions each after a runtime-only repair, with all valid controls also at zero. The next hypothesis should target action-effect inference and no-op recovery using one shared history-conditioned policy, with matched action budgets and history/blind ablations; game-ID routing or another game-specific campaign would not test that hypothesis. New-game claims require genuinely unused games, which these reports do not supply.

Across all three, establish diverse behavior, failure-informed literal mutations, joint cell/router crossover and retained gains across multiple seeds and generations under one predeclared accounting scheme. Host algorithm changes, inference-time program synthesis and authored tensor changes must retain separate provenance and controls. Official harnesses can inform protocol design, but their provider-model scores do not predict an offline bank's score. The ARC-AGI-2 evaluation exclusion remains; private or semi-private evaluation services must not become an iterative tuning channel [17]. This agenda authorizes no run, new data access, submission or expenditure by itself.

7.1 A falsifiable next shared-weight hypothesis

The completed fixed-T8 tranche in Section 5.7 did not demonstrate improvement: its founder duplicated identity and no mutation tensor was returned. The separate prompt comparison (Section 5.8) likewise leaves the explicit-objective arm unmeasured after timeout. These limited results do not answer the broader question below or authorize another call. Any subsequent study would require a new prospectively frozen protocol.

The remaining research question is whether training-only failure feedback can improve a *single reusable literal-weight artifact* while its host computation remains fixed. A proposed comparison is

$$\Delta_{\text{LLM}} = A(\theta_{\text{LLM}}, D_{\text{report}}) - A(\theta_{\text{founder}}, D_{\text{report}}),$$

against the same founder with matched random edits and against the simplest capable host-written reference. Here A is task-exact accuracy, θ includes the selected cell and router tensors, and D_{report} is frozen and unopened until all search-based selection ends. This notation defines a proposed estimand, not a measured effect. Success would require a predeclared positive margin over both the founder and matched random comparison with valid controls, plus preservation of frozen competence anchors. Matching the host reference can support successful parameter authoring, but cannot establish invention of its supplied algorithm.

Before authoring, the protocol must name a concrete expressive cell or typed bank, freeze its executable source, editable tensor vocabulary, task-family manifest, selection rule, outcome threshold, call and position caps, random seeds and stop rule. An expressibility check must first show that the requested input distinctions survive the representation: the reported `assoc_induct` equal-color invariant illustrates a case that numerical mutation cannot repair. If new structure is required, its author and control comparison belong to a separate structure experiment.

The experiment should retain every invalid and functionally duplicate proposal in accounting, show only training failures to authors, and select both candidate and random winners before final access. Independent replay must strip query labels and record valid wrong answers separately from exceptions, timeouts and malformed tensors. A new report panel would be required because the current direct and synthetic panels are exposed or spent. Acquisition and access must remain within existing permissions. The direct-study STOP and both terminal NCA stops supply no further author-call budget; this proposal starts no run.

A successful finite probe would justify testing repeated failure-driven mutation and joint cell/router crossover across seeds. It would not itself satisfy the broader `ONE_STANDARD` population requirements, demonstrate ARC-1/2/3 unification or establish general intelligence. Direct solver calls should remain a separate baseline: their different task panels, model identities/reasoning settings and budgets do not constitute a controlled causal comparison with the tensor studies.

Longer-term goals retained from the original draft. The path toward the long-term aim requires, in order: robust alphanumeric recognition across unseen font families; real-world text lines competitive with a parameter-matched gradient-trained CNN (the critical go/no-go test, since authoring effort scales with case diversity rather than data); constructed sequence priors; measured component reuse across visual tasks; constructed algorithmic reasoning with length generalization; and only then language. The largest risks are scale (capable language models have 10^8–10^{12} parameters, far beyond hand authoring, pushing toward weight *generators*) and provenance of linguistic competence. For the ARC line the owner's priorities are a valid open entry, stronger authored object, recoloring, counting and path cells tested on dense generators, and matched ablations of breeding mechanisms.

8 Reproducibility

Each result names its receipt under `runs/`, with graph and source hashes recorded in the hash-chained ledger; ledger counts were verified after retrieval. Fonts are not shipped (`CELLFORGE_FONT_DIR`), the final-evaluation server holds the private grammar and secret seeds, and the document PDFs are pinned but not redistributed. The pinned document source has 1,082 annotated

questions while the README states 1,091; we report the pinned count and have not resolved the difference. The active manuscript is `paper/cellforge.tex`; `cellforge_clean.tex` and the pre-cleanup files are historical snapshots. The undated historical PDFs predate this revision; the distinct `cellforge-20261008-r4.pdf` is the rendered output of the present source. The earlier dated revision-1, revision-2 and revision-3 PDFs and their validation records are preserved unchanged. The old `paper/check_numbers.py` targets the cleanup snapshot, so it is not validation of the current additions.

Fixed-T8 receipt binding. The local terminal package `reports/nca-tranche-terminal-20261008/SOURCE_RECEIPT_MANIFEST.json` binds 59 source and sanitized receipt files by byte size and SHA256; all entries matched during manuscript preparation. Under the preserved run in Section 5.7, `attempts/01/EVALUATION.json` contains the founder evaluation, `independent_replay/founder001/RESULT.json` records its matching replay, `attempts/02/host_capture/result.json` records the timeout, and `TERMINAL.json` records the closed, numerically incomplete comparison. The manuscript update checked these saved receipts without launching an author, replaying a scientific panel or changing frozen source, tensors, tests or floors. No public commit or replacement of an earlier published artifact is claimed for this local revision. The separate prompt-comparison execution is preserved in `runs/weights-nca-prompt-comparison-execution-20261008/`; its `TERMINAL.json` retains null B and B-minus-A values. The sanitized `reports/nca-prompt-comparison-result-20261008/SOURCE_RECEIPT_MANIFEST.json` binds 30 files, all matched during preparation of this revision. Restricted raw audit traces are excluded from that package.

Evidence and remaining facts. The October result sections link the exact experiment reports and diagnostic JSONs, which in turn identify saved runtime, raw response hashes, protocol/result hashes, controls and replay logs. The selected bank is a saved research artifact, not a qualified model release. Repository paths identify preserved evidence; source availability does not imply an independent scientific audit or benchmark submission. Newly added source-integrity citations pin the inspected commit `02beaf18`; later code fixes must cite their own commit and verification receipts. Missing facts include exact per-author credit receipts, independently captured platform contexts/isolation, missing model/reasoning metadata, qualification beyond the failing broad gates, fresh ARC transfer, multi-seed sustained evolution, and the literature pins listed above. The owner/author name and historical citation gaps remain unresolved. No secret or private evaluation data were retrieved for this revision.

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